

Y. Samuel Wang

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CURRENT POSITION	Assistant Professor <i>Cornell University</i> <i>Department of Statistics and Data Science</i>	2021 -
EDUCATION AND PREVIOUS ACADEMIC POSITIONS	Principal Researcher (Post-doc) <i>The University of Chicago, Booth School of Business</i>	2018 - 2021
	Ph.D. in Statistics <i>University of Washington</i>	2012 - 2018
	B.A. in Applied Math; Economics <i>Rice University</i>	2006 - 2010
PUBLICATIONS	1) Tao, K., Wang, Y. S., Ruppert, D. (2026+) Bayesian Analysis of Regression Discontinuity Designs with Heterogeneous Treatment Effects. <i>Forthcoming at Journal of Causal Inference</i> 2) Mingelgrin, S. and Wang, Y. S. (2026) Robust estimation of graphical models with measurement error: False discovery control and sensitivity analysis. <i>Forthcoming at UAI 2026</i> 3) Wang, Y. S., Kolar, M., and Drton, M. (2026). Confidence sets for causal orderings. <i>Journal of the American Statistical Association</i>, 121(553):690–703 4) Drton, M., Garrote-López, M., Nikov, N., Robeva, E., and Wang, Y. S. (2025). Causal discovery for linear non-Gaussian models with disjoint cycles. <i>Uncertainty in Artificial Intelligence (UAI)</i> 5) Zhao, B., Zhai, P. S., Wang, Y. S., and Kolar, M. (2024). High-dimensional functional graphical model structure learning via neighborhood selection approach. <i>Electronic Journal of Statistics</i>, 18(1):1042–1129 6) Duggan DiDominic, K. L., Shapleigh, J. P., Walter, M. T., Wang, Y. S., Reid, M. C., and Regan, J. M. (2024). Microbial diversity and gene abundance in denitrifying bioreactors: A comparison of the woodchip surface biofilm versus the interior wood matrix. <i>Journal of Environmental Quality</i>, 53(5):565–576 7) Wang, Y. S. and Drton, M. (2023). Causal discovery with unobserved confounding and non-Gaussian data. <i>Journal of Machine Learning Research</i>, 24(271):1–61 8) Wang, Y. S., Lee, C. J., West, J. D., Bergstrom, C. T., and Erosheva, E. A. (2023). Gender-based homophily in collaborations across a heterogeneous scholarly landscape. <i>Plos one</i>, 18(4):e0283106 9) Zhao, B., Wang, Y. S., and Kolar, M. (2022). Fudge: A method to estimate a functional differential graph in a high-dimensional setting. <i>Journal of Machine Learning Research</i>, 23(82):1–82 10) Wang, Y. S., Lee, S. K., Toulis, P., and Kolar, M. (2021). Robust inference for high-dimensional linear models via residual randomization. <i>International conference on machine learning (ICML)</i> 11) Wang, Y. S. and Drton, M. (2020). High-dimensional causal discovery under non-Gaussianity. <i>Biometrika</i>, 107(1):41–59	

- 12) Zhao, B., Wang, Y. S., and Kolar, M. (2019). Direct estimation of differential functional graphical models. *Advances in neural information processing systems (NeurIPS)*, 32
- 13) Chen, W., Drton, M., and Wang, Y. S. (2019). On causal discovery with an equal-variance assumption. *Biometrika*, 106(4):973–980
- 14) Drton, M., Fox, C., and Wang, Y. S. (2019). Computation of maximum likelihood estimates in cyclic structural equation models. *The Annals of Statistics*, 47(2):663–690
- 15) Varshney, A., Wang, Y. S., Watson, R. A., Noll, A., Rossi, J., Shah, P. B., Kaneko, T., Giugliano, R. P., and Adler, D. S. (2018). Abstract 12195: Relationship between transcatheter aortic valve replacement wait time and mortality in patients with symptomatic severe aortic stenosis. *Circulation*, 138(Suppl.1):A12195–A12195
- 16) Chen, Y.-C., Wang, Y. S., and Erosheva, E. A. (2018). On the use of bootstrap with variational inference: Theory, interpretation, and a two-sample test example. *The Annals of Applied Statistics*, 12(2):846–876
- 17) Wang, Y. S., Matsueda, R. L., and Erosheva, E. A. (2017). A variational EM method for mixed membership models with multivariate rank data: An analysis of public policy preferences. *The Annals of Applied Statistics*, 11(3):1452–1480
- 18) Wang, Y. S. and Drton, M. (2017). Empirical likelihood for linear structural equation models with dependent errors. *Stat*, 6(1):434–447

GRANTS AND AWARDS

NSF DMS CAREER Award (2026)
 Cornell Bowers Teaching and Advising Excellence Award (2025)
 Tom Ten Have Award, American Causal Inference Conference (2023)

PRESENTATIONS

2026
 Isaac Newton Institute, Cambridge University

2025
 Binghamton University, Department of Statistics

2024
 University of Washington, Statistics and Machine Learning for the Social Sciences
 ASA Conference on Statistical Learning and Data Science
 University of Connecticut, Department of Statistics

2023
 TU Munich, Computation, Information and Technology
 University of Texas at Austin, Salem Center for Public Policy
 New England Statistics Symposium (NESS)
 Western North American Region (WNAR) of the International Biometric Society
 Algebraic Statistics for Ecol. and Biol. Systems, Inst. for Math. and Stat. Innovation
 Computational and Methodological Statistics (CMStat)

2022
 UPSTAT
 Online Causal Inference Seminar
 Johns Hopkins Causal Inference Working Group
 Cornell Center for Applied Math

2021
 Colorado State, Statistics
 Emory, Quantitative Theory and Methods
 University of Southern California, School of Business

Cornell, Statistics and Data Science
 University of Waterloo, Statistics and Actuarial Science
 University of Toronto, Statistics
 Stevens Institute of Technology, School of Business
 Temple, School of Business
2020
 Texas A&M, Statistics
 Shiga University, International Symposium on Data Science Research and Practice
Prior to 2020
 JSM (2015, 2017, 2018, 2019)
 CMSTAT (2018)
 American Sociological Association Annual Meeting (2018)
 Wisconsin Institute for Discovery at UW-Madison (2017)
 Toyota Technical Institute - Chicago (2017)

TEACHING

Cornell University
 STSCI / INFO / ILRST 3900: Causal Inference
 STSCI 6940: Graphical Models (Special Topics)
 BTRY 6020: Statistical Methods II

University of Washington
 STAT 311: Elements of Statistical Methods

FUNDING

NSF DMS 2542851 (CAREER Award). \$450K (2026-2031)

STUDENTS SUPERVISED

PhD Thesis Advisor:
 Nabil Boulos (ongoing, joint with Marty Wells)
 Filippo Flocchi (ongoing)
 Sithija Manage (ongoing, joint with Marty Wells)
 Shira Mingelgrin (ongoing)
 Kevin Tao (ongoing, joint with David Ruppert)

PhD Thesis Committee Member:
 Daniel Coulson (ongoing, Statistics)
 Jiwoo Kim (ongoing, Psychology)
 Daniel Molitor (ongoing, Info Sci)
 Lauren San Diego (ongoing, Nutrition)
 Samantha De León Sautu (ongoing, Human Development)

 Katie Duggan (2024, Bio and Envir Engineering)
 Inbeom Lee (2024, Statistics)

Undergraduate Independent Study
 Abby Sachar (2022-23)
 Ziqing Wang (2021-22)

TECHNICAL REPORTS AND SOFTWARE

- 1) Wang, Y. S., Erosheva, E. A. (2016) “On the relationship between set-based and network-based measures of gender homophily in scholarly publications” [[arXiv](#)]
- 2) Wang, Y. S., Erosheva, E. A. (2015) “Fitting mixed membership models using `mixedmem`” [[CRAN](#)]

PROFESSIONAL
SERVICE

Journal Referee: Annals of Applied Statistics; Annals of Statistics; Bernoulli; Biometrika; Biometrics; Computational Statistics and Data Analysis; Electronic Journal of Statistics; IEEE Transactions on Neural Networks and Learning; International Journal of Approximate Reasoning; Journal of the American Statistical Association; Journal of Causal Inference; Journal of Computational and Graphical Statistics; Journal of Machine Learning Research; Journal of the Royal Statistical Society: Series B; Statistica Sinica

Conference Referee: AISTATS; ICML; NeurIPS; UAI; CLear

INDUSTRY
EXPERIENCE

Susquehanna International Group
Assistant Trader Intern

2013

Deloitte
Strategy and Operations Consultant

2010 – 2012